

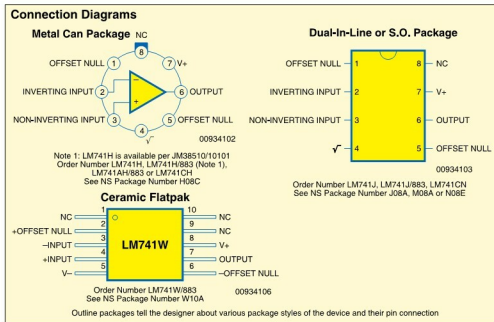


Fig. 2: Package styles of op-amp type LM741

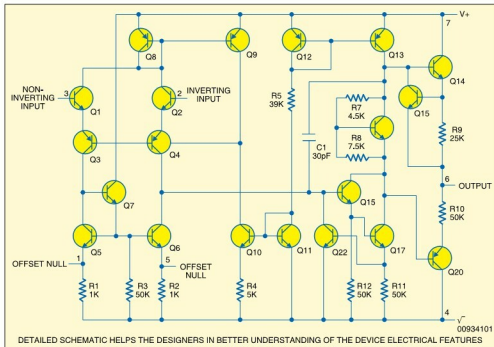


Fig. 3: Internal schematic of op-amp type LM741

Common features

A study of the datasheets of a variety of devices, be they discrete or integrated circuit (IC), reveals a commonality in the type of data and characteristic curves. These common features are outlined in the following paragraphs.

Typical applications and outstand-

ing features. All datasheets begin with typical applications as suggested by the manufacturer and the outstanding features of the device. This information appears on the first page of the datasheet and is an essential part of both condensed and detailed versions. Statements like "It has high dv/dt

capability and a low thermal resistance... designed primarily for half-wave AC power control such as motor controls, heating controls... (taken from the datasheet of S2800 series SCR)... high photosensitivity, hermetically sealed... fast optical sensor of high-modulation bandwidth... (taken from the data sheet of PIN photodiode type BPX-65)... single-pulse avalanche energy rated, nanosecond switching speeds, linear transfer characteristics, high input impedance... well suited for applications such as switching regulators, motor drivers, relay drivers... (taken from the datasheet of power MOSFET type IRF250 shown in Fig. 1)" are representative of the first page of the datasheet of the device. This gives designers a fair idea of the important features and applications without going through the whole datasheet.

Device package outline.

A sketch with device dimensions and lead identification, outline of the package styles of the device and other mechanical information are included in this part. Quite often a given device is made in more than one package style either with no difference in performance specifications (such as medium-power silicon transistor type 2N6045/MJE6045 made in TO220AB and TO127 package styles), or with a minor difference

(such as bipolar transistor 2N2222 in TO-92 plastic and TO-18 metal can packages), and occasionally with a major difference (such as IC regulator type 78-series in TO-92 plastic, TO-220 and TO-3 packages).

Fig. 2 shows the package options for the operational amplifier LM741.